SRS REVIEW 1 & 2 · PROCESS + QUALITY EVALUATION

TQM Pharm Enterprise

Process Map, Stakeholders, Measurable Requirements & Quality Goals (TQM Architecture)
Assigned Goal: Q01 Improve Reliability & Zero-Defects

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EVALUATION SCOPE:

🏛️ **Amrapali University**

B.Tech CSE Project Reviews 1 & 2
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Vision, Mission & Quality Objectives

VISION

A healthcare dispensary where zero expired, contraindicated, or unrecorded medications ever cross the counter.

MISSION

Engineer an enterprise-grade local-first operating system that automates FEFO inventory, blocks clinical contraindications (CDSS), validates narcotics, and maintains immutable audit logs.

QUALITY OBJECTIVE OF THIS SOFTWARE

Prevent errors at the source (Poka-Yoke error-proofing).

Control the process with deterministic rules. Monitor all transactions with immutable logs and real-time IoT telemetry.

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Three Quality Ideas Behind the Software

Error Prevention at Source

Poka-yoke Mistake Proofing

- Blank medicine names hard blocked.
- Qty > available stock blocked.
- Expired items blocked from cart.
- Patient allergen contraindications (Penicillin) locked.

Process Control

One Deterministic Work Path

- Supplier (GRN) → Stock (FEFO)
  → CDSS Clinical Gate → POS Bill
  → Daily Z-Report Shift Close.
- UI cannot skip or bypass business rules.

Continuous Monitoring

See, Trace & Feedback

- Live dashboard KPI alerts.
- 24/7 cold-chain IoT logs (2-8°C).
- 21 CFR Part 11 immutable audit trail.
- Error logs feeding FMEA & Pareto.

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Process Map & Controls Across the Pipeline

Supplier → Purchase (GRN) → Stock (FEFO) → Bill (POS) → Customer / Fleet → Z-Report Close

Controls sitting on this process (Quality Gates)

On Inwarding (GRN)
Batch lot, quantity, cost price, expiry must be filled. Dockside supplier verification against PO.

On Stock Ledger
FEFO dynamic sorting; low-stock thresholds (≤20); 30/90-day expiry tags; 2-8°C IoT temperature alarms.

On POS Billing
CDSS allergy cross-check; expired lot hard lockout; atomic transaction stock deduction; audit logging.

On Shift Close (Report)
Daily Z-Report mathematical cash balancing; till variance detection; error logs feeding FMEA.

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SIPOC & Stakeholder Requirements Matrix

📊 SIPOC OF THE DISPENSING PROCESS

Suppliers: Pharmaceutical distributors, attending doctors, SQLite database.

Inputs: e-Prescription, batch lot, dispense qty, patient allergy EHR, cashier role.

Process: Validate role → CDSS scan → FEFO pick → Deduct stock → Save + Audit.

Outputs: Official GST invoice, thermal receipt, decremented stock, delivery order.

Customers: Patient, cashier counter, store owner, state drug inspectors.

👥 STAKEHOLDERS & CORE NEEDS

Patient: Safe medication without allergens; transparent GST billing.

Cashier: Fast sub-second billing; automated change return & dynamic UPI QR.

Pharmacist: Real-time stock; CDSS allergy alerts; Schedule X narcotic register.

Admin / Owner: Staff RBAC; live terminal presence; shift Z-Report cash balancing.

Auditor / Inspector: 21 CFR Part 11 immutable audit trail & 1-click narcotic CSV export.

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Measurable Requirements & Assigned Goal Q01

🎯 MEASURABLE REQUIREMENTS (VIVA)

No Expired Sale: Expired sales in a month = 0 (100% hard block at POS cart).

Stock is True: Physical inventory variance < 1.5% across all SKUs.

No Oversell: Qty > stock attempts blocked = 100% via atomic assertion.

Trace Every Change: 100% of stock/price edits have user ID + timestamp in audit_logs.

Fast Search: Barcode & generic search latency < 50ms (exceeds <3s target).

Early Risk Flag: Items ≤30d (red) and ≤90d (amber) automatically tagged on UI.

🏆 ASSIGNED QUALITY GOAL Q01: IMPROVE RELIABILITY

1. Auto Backup: Automated snapshot copy of pharmacy_system.db created daily in backups/.

2. Audit Log: Captures who, action, old value, new value, terminal IP, and ISO timestamp.

3. Input Validation: Stops blank names, negative prices, oversell quantities, bad dates.

4. Error Recovery: Atomic rollback on commit failure; no half-created bill; friendly toasts.

5. User Roles (RBAC): Store Admin, Pharmacist, Cashier, Inventory, Delivery strictly isolated.

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Modular 6-Tier Architecture & The Golden Rule

🗂️ 6-TIER FOLDER ARCHITECTURE

```
Pharmacy_Management_System/
├── core/
│   ├── bridge.py       # RPC Bridge (38 typed methods)
│   ├── db.py           # SQLite persistence (14 tables)
│   ├── api_client.py   # Telemetry & presence client
│   └── logger.py       # 21 CFR Part 11 audit logger
├── config/
│   └── settings.py     # Central Singleton configuration
├── ui/
│   ├── desktop/        # Edge Chromium WebView2 60 FPS
│   └── web/            # Cloud-ready FastAPI Web UI
├── apps/               # Admin & operational workstations
├── tests/              # 24 Automated pytest test cases
├── backups/            # Automated SQLite snapshots
├── main.py             # Hardware-accelerated entrypoint
└── web_app.py          # Asynchronous FastAPI web server
```

⭐ THE ONE RULE TO REMEMBER

"The screen must not talk to the database."

UI → Service (Bridge) → Model / Database

- No SQL queries inside billing.py or any UI template.
- All database operations are mediated by typed Python methods.
- Failure isolation: UI bugs cannot corrupt database ACID commits.
- Modularity allows independent testing and zero regression.

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How a Click Travels (The 7-Step Architectural Pipeline)

1. Screen (UI Layer)
billing.py captures user click, requested medicine SKU, and dosage quantity from input/barcode.

2. Service (Bridge Layer)
sales_service checks role authorization, verifies terminal session, and initiates transaction.

3. CDSS Engine (Clinical Layer)
medicine_service cross-checks patient EHR for penicillin/NSAID allergies and drug interactions.

4. Utils (Validators Layer)
validators.py rejects bad inputs: checks non-blank, non-negative, and quantity <= available stock.

5. Database (Storage Layer)
db.py executes single ACID transaction: writes sales invoice and atomically cuts stock quantity.

6. Proof (Audit Layer)
audit_logs and error_logs receive append-only timestamped records containing user ID and payload.

7. Later (TQM Analytics Layer)
tqm_report, Pareto charts, and FMEA tables aggregate logs to monitor and eliminate recurring defects.

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Database Design & Relational Schema (14 Tables)

CORE RELATIONAL ENTITIES

- **1. staff:** Credentials, assigned RBAC role, branch, active status.
- **2. medicines:** SKU, generic, batch, expiry, days left, stock, MRP, rack.
- **3. prescriptions:** Patient name, doctor, items JSON, CDSS status, clinical notes.
- **4. invoices:** Fiscal tax invoices, cashier, subtotal, GST split, total.
- **5. delivery_orders:** Dispatch route, address, rider, cold-bag check, 4-digit OTP.
- **6. patients:** EHR profiles, contact phone, chronic diseases, recorded allergies.
- **7. suppliers:** Distributor directory, GSTIN, contact phone, payment terms.

COMPLIANCE & TELEMETRY ENTITIES

- **8. purchase_orders:** GRN inwarding, supplier, batch items JSON, total amount.
- **9. narcotic_register:** Schedule X: Doctor DMC Reg No, Pharmacist PIN, quantity.
- **10. temperature_logs:** 24/7 cold-chain telemetry: probe ID, temperature (2-8°C).
- **11. refill_schedules:** Chronic maintenance therapy: last dispensed, next due date.
- **12. active_sessions:** Live presence: user ID, terminal name, 5s heartbeat timestamp.
- **13. audit_logs:** 21 CFR Part 11 append-only log: user, role, action, details.
- **14. z_reports:** Daily till close: opening float, sales, payouts, live variance.

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Role-Based CRUD Operations Matrix

Module / Entity	👑 Store Admin	🩺 Pharmacist	💳 Cashier	📦 Inventory	🛵 Delivery
Staff & RBAC	CRUD	R	R	R	R
Medicines & Stock	CRUD	R	R, U (Deduct)	CRUD	R
Prescriptions Queue	R	CRUD (Validate)	R, U (Dispensed)	R	R
Invoices & POS	R	R	CR (Bill & Print)	R	R
Delivery Orders	R, D	R	C (Dispatch)	R	RU (OTP Verify)
Narcotic Register	R (Export CSV)	CR (Double Sign)	R	R	No Access
Daily Z-Reports	R, U (Audit Sign)	No Access	CR (Shift Close)	No Access	No Access

Software Features → TQM Concept Mapping

Implemented Software Feature	Academic TQM Quality Principle / Concept
Modular Architecture (6-Tier)	System Approach / Process Organization (Deming)
CDSS Allergy & Drug-Drug Shield	Poka-Yoke Error Proofing at Source (Shingo)
FEFO Batch Sequencing Algorithm	Process Control & Inventory Wastage Elimination
24/7 Cold-Chain IoT (2-8°C)	Quality Assurance & Biological Preservation
Daily Z-Report Till Reconciliation	Fact-Based Management & Fiscal Governance (Juran)
Immutable Audit Logs (21 CFR Part 11)	Traceability, Accountability & Non-Repudiation
Automated Error Logs & Telemetry	Root-Cause Prioritization (Pareto 80/20) & FMEA
Customer / Patient EHR Profiles	Customer / Patient Focus (Zero-Defects Philosophy)

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Quality Dashboard & Operational Benchmarks

Stock Accuracy
99.2%
Zero unrecorded shrinkage

Billing Errors
0.0%
Poka-Yoke blocks bad data

Expired Sales
0.0%
100% FEFO cart lockout

Till Variance
₹ 0.00
Z-Report cash balanced

📊 TRADITIONAL PHARMACY VS. TQM PHARM TQM BENCHMARK

Quality Dimension	Traditional Retail Pharmacy (Manual)	TQM Pharm Enterprise (TQM Driven)
Prescription Safety	Manual check; vulnerable to handwriting errors	Automated CDSS cross-checks with patient allergy EHR
Inventory Picking	Random or FIFO; older batches expire on shelves	Algorithmic FEFO (First-Expiry, First-Out) guarantee
Cold-Chain Monitoring	Infrequent manual thermometer checks; silent loss	Continuous 30s IoT telemetry with automatic alarm alerts
Shift Cash Balancing	Paper ledger estimations; undetected shrinkage	Daily Z-Report mathematical reconciliation & signed Z-slip

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Quality Analysis Tools: Pareto (80/20) & Fishbone

📈 PARETO ROOT-CAUSE ANALYSIS (80/20 RULE)

1. Look-Alike / Sound-Alike Drugs (45%): Addressed by CDSS generic molecule search and allergy confirmation.

2. Qty > Available Stock Attempts (30%): Addressed by pre-save inventory balance assertion locks.

3. Near-Expiry Drug Dispensing (15%): Addressed by algorithmic FEFO nearest-expiry prioritization.

4. Insulated Cold-Bag Packaging (7%): Addressed by mandatory delivery pre-dispatch checklist.

Key TQM Takeaway: Eliminating the top 2 error modes removes 75% of all operational dispensary defects.

🐟 FISHBONE (ISHIKAWA) ROOT-CAUSE BREAKDOWN

People (Staff): Rushed cashiers during counter peaks; new dispensers misinterpreting doctor handwriting.

Process (Workflow): Lack of second-check clinical verification step before POS billing.

Software (System): Legacy systems allowing expired stock cart addition and lacking allergy alerts.

Data (Integrity): Missing batch numbers, incorrect physical expiry dates, unrecorded patient allergies.

Machine (Hardware): Slow workstation hardware and missed clicks on crowded UI screens.

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FMEA & How Telemetry Logs Feed Risk Reduction

⚠️ FMEA TABLE (RPN = S x O x D)

Sell Expired Drug: S=10, O=4, D=3 → RPN 120

Action: Hard block prevents cart checkout.

Dispense Allergen Drug: S=10, O=3, D=4 → RPN 120

Action: CDSS allergy warning & override block.

Qty > Stock Oversell: S=8, O=5, D=3 → RPN 120

Action: Pre-commit inventory balance assert.

Duplicate Barcode Scan: S=6, O=2, D=2 → RPN 24

Action: Unique constraint on SKU barcode.

🔄 HOW ERROR LOGS FEED FMEA & IMPROVEMENT

1. Attempt Blocked: Cashier attempts bad action or system encounters exception.

2. Log Appended: error_logs and audit_logs record error code (E01-E05) + user.

3. Aggregation: tqm_report counts monthly frequency of each error code.

4. Dynamic Occurrence: High frequency automatically raises Occurrence (O) score in FMEA.

5. Prioritized Action: High RPN items trigger software updates and staff retraining.

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Deming PDCA Cycle & Transaction Workflows

🔁 DEMING PDCA QUALITY CYCLE

PLAN: Identify excessive oversell and near-expiry mistakes during peak rush.

DO: Deploy automated FEFO batch picker and pre-save stock balance locks.

CHECK: Monitor monthly counts of error codes E01 and E03 in error_logs.

ACT: Standardize FEFO logic across all terminals as mandatory baseline policy.

SALE & PURCHASE WORKFLOW SEQUENCES

Dispensing Sale Flow: Login → POS Billing → Barcode / 1-Click eRx → Qty → Validate → Stock Check → Expiry Check → ACID Commit & Thermal Receipt.

Consignment Purchase Flow: Supplier Pick → Inwarding Modal (GRN) → Batch & Expiry Entry → Validate → Stock Increment → Reorder Evaluation → Telemetry Alert.

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Working Application Demo Guide: Setup & Personas

🚀 APPLICATION LAUNCH COMMANDS

Automated Tests (Quality Proof): `pytest`

→ 24/24 unit & integration tests passing in ~7.8s.

Desktop Workstation App: `python main.py`

→ Boots hardware-accelerated Edge Chromium WebView2 window.

Cloud Web Portal: `python web_app.py`

→ Starts FastAPI server at http://127.0.0.1:8000 with Swagger docs.

Terminal Binding: Assign session to Front Counter A, Clinical Kiosk, or Dock.

🔑 WORKSTATION PERSONAS & CREDENTIALS

👑 **Admin: `admin` / `admin123`**

Scope: RBAC, live heartbeats, Z-Report balancing

🩺 **Pharmacist: `pharmacist` / `pharma123`**

Scope: CDSS allergy scan, Schedule X narcotic sign-off

💳 **Cashier: `cashier` / `cash123`**

Scope: 1-Click eRx import, FEFO batch, UPI QR, receipt

📦 **Inventory: `inventory` / `inv123`**

Scope: Multi-batch ledger, GRN inwarding, 2-8°C IoT

🛵 **Delivery: `delivery` / `deliv123`**

Scope: Kanban dispatch, cold-bag check, OTP e-POD

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Live Demonstration Workflow: Quality Gates in Action

Phase 1: Clinical Pharmacist & CDSS Gate

Review incoming eRx queue. Trigger CDSS Allergy Interception (Penicillin warning blocks checkout). Execute Schedule X Narcotic double sign-off with Doctor DMC No + Pharmacist PIN.

Phase 2: POS Cashier Express Billing

1-Click import approved eRx into cart. Verify automatic FEFO nearest-expiry batch selection. Tender checkout: enter Cash (auto change calc) or Dynamic UPI QR. Print 80mm GST receipt.

Phase 3: Inventory FEFO & Cold-Chain IoT

Inspect batch ledger with countdown badges (<30d Red, <90d Yellow). Click 'Simulate Temp Spike' (9.4°C) on Chamber A fridge → observe immediate visual alert & log entry.

Phase 4: Fulfilment Fleet & OTP Handover

Advance order on Kanban dispatch board. Verify insulated cold-bag checklist. Enter customer 4-digit SMS OTP (1234) to confirm electronic Proof-of-Delivery (e-POD).

Phase 5: Admin Command Center & Z-Report Close

Monitor live terminal presence heartbeats. Open Daily Z-Report module: enter physical cash counted, verify variance (Balanced ₹0.00), sign and lock shift register.

REVIEW 1 VS REVIEW 2 ROADMAP & VIVA CONCLUSION

Review 1 Delivered vs Review 2 Horizon

Review 1 Delivered (Now): Complete 6-tier architecture, IEEE 830 SRS, 14 tables, working desktop & web UIs, CDSS allergy interception, FEFO batch dispensing, Daily Z-Report till balancing, 24/24 passing tests.

Review 2 Scope (Horizon): Longitudinal FMEA occurrence tracking from error logs, automated real-time Pareto charts, Ishikawa incident registry, and multi-branch cloud database replication.

⭐ ONE SENTENCE FOR VIVA CLOSING

"The UI never writes SQL. Validation stops bad data. Audit records who did what. Expired stock cannot be sold. Logs become Pareto, FMEA, PDCA."

🙏 Sincere gratitude to the Computer Science & Management faculty at Amrapali University. The floor is open for viva questions. (Candidate: Piyush Bhatt)